

CHANG LIU

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EDUCATION

Texas A&M University

Ph.D., Mechanical Engineering – Advisor: Dr. Minghui Zheng

College Station, TX

Jan. 2024 – Present

University at Buffalo, SUNY

M.S., Mechanical Engineering – Advisor: Dr. Minghui Zheng

• Thesis: *Deep Learning for Collaborative Robot Safety.*

Buffalo, NY

Aug. 2018 – Feb. 2021

Huazhong University of Science and Technology

B.S., Energy and Power Engineering

Wuhan, China

Sep. 2014 – June 2018

RESEARCH INTERESTS

My research focuses on robot learning and perception for contact-rich robotic disassembly, with the long-term goal of raising the automation level of remanufacturing. I build visual perception pipelines (YOLO, GNN, vision-language models) that guide disassembly planning under uncertainty, and I develop vision-language-action (VLA) models that turn those plans into reliable manipulation policies on real hardware. Specific interests include VLM-based reasoning under uncertainty, skill-based agentic VLA frameworks, reinforcement learning from human corrections, and multimodal data collection pipelines for foundation-model training.

PUBLICATIONS

Author name in **bold** indicates first-author contribution. Full list also available on [Google Scholar](#).

Peer-Reviewed Journal Articles

- [1] **C. Liu**, S. Behdad, P. Pagilla, X. Liang, M. Zheng. “Transforming Remanufacturing Automation with Large Language Models: A Forward-Looking Analysis with Case Studies.” *Robotics and Computer-Integrated Manufacturing* 103 (2027), 103361.
- [2] X. Zhang, **C. Liu**, X. Liang, M. Zheng, S. Behdad. “Evaluating Large and Lightweight Vision Models for Irregular Component Segmentation in E-Waste Disassembly.” *Journal of Manufacturing Science and Engineering* 148(10) (2026), 101005.
- [3] **C. Liu**, B. Balasubramaniam, N. A. Yancey, M. H. Severson, A. Shine, P. Bove, B. Li, X. Liang, M. Zheng. “RAISE: A Robot-Assisted Selective Disassembly and Sorting System for End-of-Life Phones.” *Resources, Conservation and Recycling* 225 (2026), 108609.
- [4] H. Zhao, **C. Liu**, B. Balasubramaniam, J. Li, J. Song, X. Liang, M. Zheng, B. Li. “Precision 3D Profilometry of Consumer-Grade Computer Enclosures Using High Dynamic Range Fringe Projection.” *Applied Optics* 64(30) (2025), 8986–8994.
- [5] W. Liu, **C. Liu**, X. Liang, M. Zheng. “A Hybrid Task-Constrained Motion Planning for Collaborative Robots in Intelligent Remanufacturing.” *Mechatronics* 102 (2024), 103222.
- [6] W. Liu, **C. Liu**, S. Sajedi, H. Su, X. Liang, M. Zheng. “An Audio-Based Risky Flight Detection Framework for Quadrotors.” *IET Cyber-Systems and Robotics* 6(1) (2024), e12105.
- [7] Z. Chen, **C. Liu**, H. Su, X. Liang, M. Zheng. “A Hybrid Disturbance Observer for Delivery Drone’s Oscillation Suppression.” *Mechatronics* 88 (2022), 102907.

Peer-Reviewed Conference Papers

- [8] C. Liu, et al. “PACER: Process-Aware VLA Policy Learning from Human Corrections for Robotic Disassembly.” *Conference on Robot Learning (CoRL)*, 2026. (Accepted)
- [9] C. Liu, W. Liu, Z. Chen, M. Zheng. “A Deep-ConvLSTM Collision Prediction Model for Manipulators in Dynamic Environment.” *IFAC-PapersOnLine* 55(27) (2022), 68–75.

Preprints and Manuscripts Under Review

- [10] C. Liu, S. Tian, X. Liang, M. Zheng. “SELF-VLA: A Skill-Enhanced Agentic Vision-Language-Action Framework for Contact-Rich Disassembly.” *arXiv preprint arXiv:2603.11080* (2026).
- [11] S. Tian, C. Liu, M. Zheng, X. Liang. “NeurRAFT: Robot Motion Planning via Anchor-Level Flow Matching with Clearance-Aware Preference Tuning.” *arXiv preprint arXiv:2608.24026* (2026).
- [12] J. Lei, C. Liu, Y. She, X. Liang, M. Zheng. “Learning When to See and When to Feel: Adaptive Vision-Torque Fusion for Contact-Aware Manipulation.” *arXiv preprint arXiv:2604.01414* (2026).
- [13] C. Liu, S. Tian, S. Behdad, X. Liang, M. Zheng. “Vision-Language-Action Models for Selective Robotic Disassembly: A Case Study on Critical Component Extraction from Desktops.” *arXiv preprint arXiv:2512.04446* (2025).

RESEARCH EXPERIENCE

Graduate Research Assistant

Control and Robotics Lab, Texas A&M University – Advisor: Dr. Minghui Zheng

College Station, TX

Jan. 2024 – Present

SELF-VLA: Skill-Enhanced Agentic Vision-Language-Action Framework

Jun. 2024 – Present

- Built an agentic skill-based VLA framework that decomposes contact-rich disassembly into learnable manipulation skills and uses a vision-language model as a high-level reasoner to handle scene ambiguity and task uncertainty.
- Fine-tuned and deployed OpenVLA, OpenVLA-OFT, and OpenPI policies on a UR5e robot, improving task success rate by 40%+ over end-to-end VLA baselines.
- Designed a process-aware correction-based training loop (**PACER**, accepted to CoRL 2026): when a skill fails on the real robot, the operator issues a short teleoperated intervention and the correction is folded back into the policy via reinforcement learning with human feedback.

RAISE: Robot-Assisted Disassembly and Sorting for End-of-Life Products

Jan. 2024 – Present

- Built an end-to-end robotic disassembly and sorting system for end-of-life phones in collaboration with Idaho National Laboratory: hand-eye calibration on a UR5e, YOLO-based component detection at 99%+ mAP, and automated sorting of screws, batteries, and circuit boards.
- Integrated perception, sequence planning, and manipulation into a single autonomous pipeline, covering multiple phone models without operator intervention.

Perception Pipeline and Data Infrastructure for Disassembly

Jan. 2024 – Present

- Built a real-time perception pipeline combining YOLO detection, GNN-based disassembly sequence reasoning, and VLM-guided decision-making for end-of-life desktop computers, enabling autonomous part selection under partial occlusion and unseen device variants.
- Built and open-sourced a teleoperation and data collection platform on the GELLO framework: ZMQ-based UR5e control, dual-camera streaming (RealSense D435i + OAK-D Pro at 30 Hz), HOSAS 6-DOF joystick, and phase-labeled episode recording for OpenVLA / OpenVLA-OFT / OpenPI training.
- Collected 500+ teleoperated demonstrations and a 20,000+ image multimodal dataset across 15+ device variants, annotated for object detection, segmentation, and graph-based disassembly reasoning.

Graduate Research Assistant

CAL-UB Lab, University at Buffalo, SUNY – Advisor: Dr. Minghui Zheng

Buffalo, NY

Aug. 2019 – Apr. 2021

- Built a Deep-ConvLSTM model for vision-only collision prediction in collaborative robot workspaces, reaching 95% accuracy without access to robot dynamics.
- Applied SAC reinforcement learning for safety-aware motion planning in shared human-robot workspaces.
- Built a transfer-learning pipeline for quadrotor propeller fault detection from audio spectrograms, reaching 98% accuracy across 4 fault types.

INDUSTRY EXPERIENCE

Senior Technical Support Engineer

Irvine, CA

HeyGears Inc. – Industrial 3D Printing

Apr. 2021 – Nov. 2023

- Managed 70+ clients and 300+ industrial 3D printers across the East Coast; ran on-site product testing and wrote technical reports that drove firmware and hardware fixes.

TECHNICAL SKILLS

Perception & Vision: YOLO, Segmentation, GNN-based Scene Reasoning, Vision-Language Models, Multi-view Fusion, RealSense D435i, OAK-D Pro

Robot Learning: VLA (OpenVLA, OpenVLA-OFT, OpenPI), Imitation Learning, Reinforcement Learning (SAC, PPO), Transformers, Flow Matching, Foundation Model Fine-tuning

Robotics: ROS2, MoveIt, Isaac Sim, MuJoCo, UR5e, Teleoperation (GELLO, HOSAS), Vicon, Hand-Eye Calibration

Languages: Python, MATLAB, SQL

Tools: PyTorch, TensorFlow, Git, Docker, Linux, CUDA, ZMQ